

A computational fluid dynamics study by conjugate heat transfer in OpenFOAM: A liquid cooling concept for high power electronics

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ABSTRACT

A conjugate heat transfer (CHT) study of a liquid cooling heat exchanger is carried out using the open source computational fluid dynamics (CFD) library OpenFOAM. The heat exchanger was 3D printed using aluminium and experimentally verified by temperature probing and thermal imaging. The functionality of the heat exchanger in cooling localized heat sources is demonstrated. Three different turbulence models were utilized including $k-\omega$ shear stress transport (SST) model, the standard $k-\epsilon$ model and large-eddy simulation (LES). The numerical results indicate that the $k-\omega$ SST and LES models produced similar results in terms of flow structures and temperature levels while the $k-\epsilon$ model deviated from the two other models. The scalability of the heat exchanger was numerically demonstrated by comparing the flow uniformity by varying the inlet Reynolds number between 4960 and 14880. The conclusions of the paper consists of the following main results. (1) The numerical results indicate that the flow uniformity in the channels is noted to be affected by the flow structures before and after the fin system. (2) The simulated hot-spot temperatures were noted to be relatively sensitive to the predicted flow laminarization inside the channels. (3) The heat exchanger was shown to be functional and to maintain cool surface temperatures in the simulations and the experiments. Additionally, the used CHT solver in OpenFOAM is tested and verified in different ways.

1. Introduction

Along with electrification and digitalisation, the societal demand for liquid cooling as an alternative or complementary approach for air cooling is constantly increasing. In comparison to air, water has clearly higher heat capacity $c_{p,w}/c_{p,a} \sim 4$, substantially higher heat conductivity $\kappa_w/\kappa_a \sim 30$ and it is three orders of magnitude higher in density $\rho_w/\rho_a \sim 1000$ than air. These aspects motivate the usage of water as coolant with obvious potential for more compact cooling concept design. In this paper, the results of a compact liquid cooling solution for high power electronics are reported.

In the context of medium to high power electronics (here: $\approx 8 \text{ W/cm}^2$), a common liquid cooling heat exchanger design involves embedded microchannels, first demonstrated experimentally by Tuckerman and Pease (1981). Harms et al. (1999) compared a multi-channel design for developing laminar flow to a single channel design for a turbulent flow. The multi-channel design was noted to outperform the single channel design due to the increased heat transfer rate in the entrance region. Qu et al. (2000) studied the heat transfer in trapezoidal silicon microchannels experimentally and compared the results

to numerical conjugate heat transfer (CHT) analysis. The authors suggested that the discrepancy could rise from failure to model surface roughness in the CHT analysis. Qu and Mudawar (2002a,b) studied microchannel heat sinks numerically with CHT in addition to performing experiments. Higher heat transfer rates were noticed in the entrance region of a developing laminar flow. Increasing the Reynolds numbers increased the length of the entrance region at the expense of higher pressure losses. Exceptional agreement between the numerical predictions and the experiments performed, was reported.

Chiu et al. (2011) studied the effect of the aspect ratio of the microchannels on the heat transfer performance numerically using CFD and experimentally. The entrance effect on the increase in the Nusselt number was observed to be higher with lower aspect ratio microchannels. Sharma et al. (2015) studied hot-spot targeted liquid cooling with manifold layers targeted on top of the hot-spots. Up to 61% lower temperature non-uniformity was achieved with a 150 W/cm^2 heat flux. Al-Waaly et al. (2017) studied experimentally and numerically, the cooling of an electronic circuit using non-uniform heat flux with additional sub-channels at the hot-spots at laminar conditions using CHT

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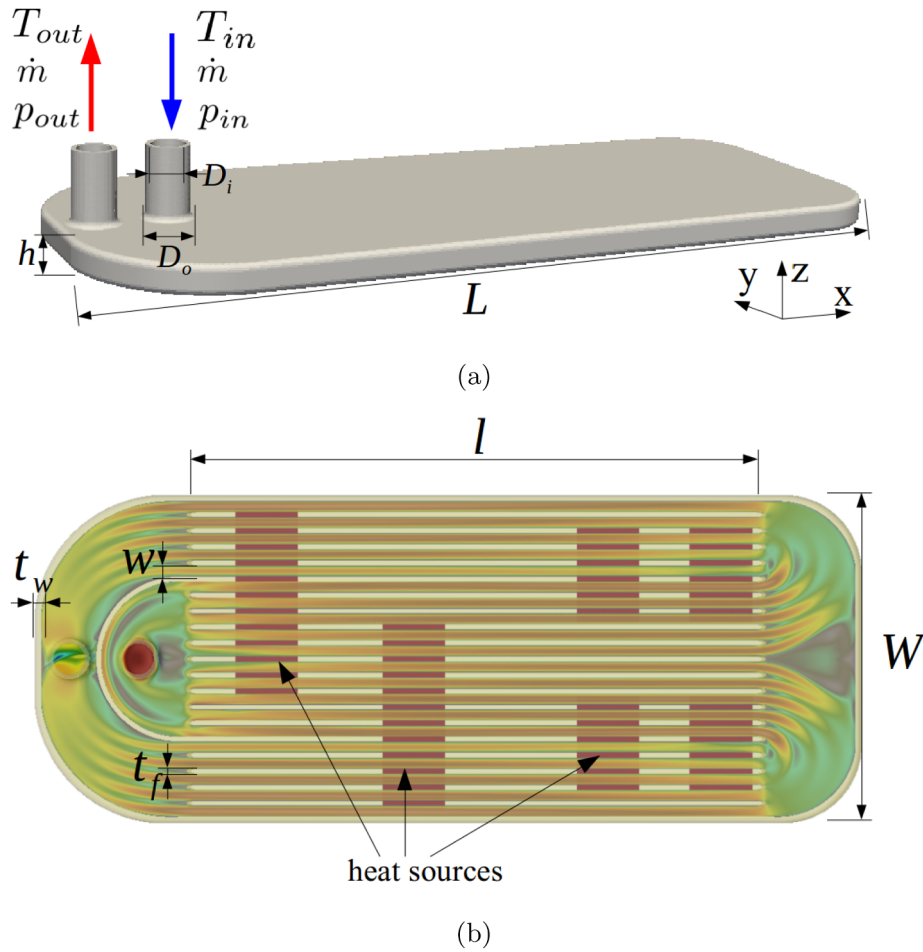


Fig. 1. (a): The designed liquid cooling heat exchanger. L : the length of the heat exchanger, D_i : inlet/outlet inner diameter, D_o : inlet/outlet outer diameter, h : height of the heat exchanger, W : width of the heat exchanger. (b): A cut-plane at the centre of the design with a simulated flow field and the locations of the heat sources. l : length of the heat transfer fins, w : width of a heat transfer channel, t_f : thickness of a heat transfer fin, t_w : wall thickness.

and CFD. Implementing additional local sub-channels reduced the temperature non-uniformity and maximum surface temperatures. [Ansari and Kim \(2018\)](#) studied a hybrid design of microchannels with pin-fins located at the hot-spots with a laminar CHT model using CFD. The implementation of the pin-fins decreased the temperature levels at the hot-spot at the same Reynolds number with a slight increase in the pressure drop. [Soleimanikutanaei et al. \(2018\)](#) studied numerically with CHT a microchannel heat exchanger including transverse microchannels. The Nusselt number was noted to increase with the total number and smaller height of the transverse microchannels due to the entrance effect. [Alfieri et al. \(2010, 2012\)](#) developed and validated a 3D CHT CFD liquid cooling model of multilayer electronic chips using a microchannel design with pin-fins on the hot-spots. The implementation of the pin-fins on the hot-spots revealed up to 20% improvement in the heat transfer rate.

[Flageul et al. \(2015\)](#) studied different boundary condition types, including conjugate and non-conjugate methods, for heat transfer in a turbulent channel flow using direct numerical simulation (DNS). The results indicated the existence of very large scale thermal structures that are intrinsic to conjugate heat transfer. In CHT, the dissipation rate involved in the temperature fluctuations is discontinuous at the solid-fluid interface. This is a consequence of discontinuous material properties across the boundary while the temperature and heat flux are continuous. It is noted that there appears to be no Reynolds-averaged Navier-Stokes (RANS) models currently that would model this discontinuity ([Craft et al., 2010; Flageul et al., 2017, 2019](#)). However, here a modeling choice was made not to include this effect in the steady state

RANS simulations.

The present study is motivated by the development of a cooling application in 5G context. In the same context, [Peltonen et al. \(2019\)](#) studied air cooling numerically using large-eddy simulation (LES). By comparing the usage of rectangular fins compared to pin fins, the results indicated a better heat transfer performance with pin fins due to vortex shedding. The simulation results were compared with experiments in terms of flow field statistics. The results indicated agreement between LES and the experiments. Also, Peltonen et al. 3D printed the experimentally tested heat exchangers using aluminium. The heat transfer approach was based on using constant fixed temperature at the surfaces of the fins. The constant temperature approach does not take into consideration the solid conduction; therefore, the overall thermal performance depends on the boundary conditions and vice versa. Peltonen et al. noted that accurate predictions of heat transfer would require a conjugate heat transfer simulation with indicated power density. The present study continues the work by Peltonen et al. Here, a more compact liquid cooling heat exchanger is designed and its heat transfer performance is analysed by performing CHT simulations. However, it is clear that microchannels as a concept is an existing technology. Yet, the detailed geometry of the present design has not been previously published as such. Furthermore, based on the literature survey, it is noted that a CHT-LES combination has not been commonly used in liquid cooling studies under turbulent conditions. Thereby, there are clear gaps in the CFD-CHT literature on microchannels in turbulent conditions. To fill the research gaps, this study has the following main objectives:

1. Verify the functionality of the CHT code used herein for laminar and turbulent flow cases.
2. Examine the fluid dynamical and thermal phenomena occurring in the liquid cooling heat exchanger with CFD-CHT.
3. Assess different turbulence models in predicting the performance of the present heat exchanger.
4. Validate the functionality of the 3D printed aluminium heat exchanger both experimentally and numerically.

2. Experimental set-up

2.1. Geometric configuration

In this study, a heat exchanger based on liquid cooling is investigated. The present geometric configurations is based on the constraints of the underlying cooling application in 5G context. The novelty of this study is in the analysis of the turbulent fluid dynamical structures using a CHT approach with CFD and comparing the overall performance to a 3D printed heat exchanger. The motivation for liquid cooling arises from the previous study by Peltonen et al. where air cooling with similar power ranges led to larger heat exchanger dimensions. The liquid cooling heat exchanger analysed in this study can be seen in Fig. 1. The heat exchanger contains twenty coolant channels for fluid flow. This increases the heat transfer surface area by adding additional walls between the channels. To reduce maldistribution of the flow rate between the channels, the inflow and outflow are placed perpendicular to the cooling plate, i.e., there is no initial velocity component in the planar direction of the cooling plate. The location of the outlet is chosen to be close to the inlet to decrease the space needed for the water connections.

The detailed dimensions of the heat exchanger are listed in Table 1. The channels are of millimetre size rather than micrometre for two reasons. Firstly, decreasing the width of the channels, and thus increasing the number of channels, would increase the pressure drop over the heat exchanger. Secondly, 3D printing small channels has certain challenges of which perhaps the most obvious one is the challenge to remove the excess metal powder from the system after the printing is complete (Edgar and Tint, 2015). The height of the coolant channels in the heat exchanger is 5 mm and the thickness of the cooling plate is uniform throughout. Geometrical constraints and limitations of the cooling application were the main design drivers for the present heat exchanger.

2.2. Experimental procedure

The functionality of the designed heat exchanger was measured experimentally by attaching an electronic board to the bottom of the heat exchanger and connecting the heat exchanger to a cooling water circuit. The electronic board was attached firmly to the 3D printed heat exchanger with thermal paste at the interface between the electronic components and the heat exchanger to ensure proper contact. The experimental set-up can be seen in Fig. 2.

The experiments were carried out with flow rates of 2, 3, 4, 5 and 6 L/min. The flow rates were measured with a Kytölä C-4BA flow meter. The voltage and electric resistance of the electronic circuit was measured with a Fluke 73 series multimeter. The test electronic circuit used in the experiments consisted of 16 TEH100 series resistors that provided heat. The resistors were attached to the heat exchanger with

Thermal Grizzly Aeronaut thermal paste, thermal conductivity of which is $\kappa = 8.5 \text{ W/mK}$. The test circuit is shown in Fig. 2b. The 16 electric resistors, i.e. the heat sources, are shown in the figure along with the cables for the thermocouples. The locations for the temperature measurements with the thermocouples are marked with numbers 1–8. The temperature levels of the electronic components were measured with K-type thermocouples. Thermoelements were calibrated mainly for temperature difference measurements. Calibration was done using a thermal bath with a temperature level of 23°C . Temperatures of all 12 thermoelements were measured three times. The standard deviations of the temperature readings were $0.0829\ldots 0.111^\circ\text{C}$ which correspond to average standard deviation of 0.0927°C . According to Student-distribution of 12 readings with a risk level of 0.05 this corresponds to an error limit of $\pm 0.06^\circ\text{C}$ for one temperature reading and $\pm 0.08^\circ\text{C}$ for a temperature difference.

Thermal images of the electronic circuit were taken with a FLIR SC7000 infrared camera. The emissivity of the electronic board outer surface (from which the thermal images were taken) was then measured. First, the outer surface of the heat exchanger and electronic circuit was painted black. The cooling water temperature was then measured using K-type thermoelements at three different constant temperature levels of 25°C , 35°C and 45°C . The aluminium heat exchanger surface temperature was simultaneously measured using the infrared camera with different emissivities ($\varepsilon = 0.9\ldots 1.0$). The measured cooling water temperatures were compared to the measured infrared camera values. Finally, the emissivity was found to be $\varepsilon = 0.96$.

The configuration of the heat sources, in addition to the total heating power, was consistent with the simulations. However, there are a few experimental details that can not be included in the simulations. Firstly, the actual temperature levels of the heated surface of the heat exchanger could not be measured with an infrared camera since the electronic circuit was firmly attached to the heat exchanger. Hence, the thermal images were taken from the back of the electronic circuit, which was not attached to the heat exchanger from parts other than the heat sources. Secondly, the thermocouples could not be located exactly at the interface between the heated component and the heat exchanger. This would have created a small air gap at the interface, resulting in a thin layer of insulation. Thirdly, the components were attached to the heat exchanger with thermal paste, which created additional thermal resistance, resulting in an increase in the measured temperatures compared to the simulations. Finally, the fluid dynamical structures of the flow field could not be captured experimentally from the inside of the aluminium heat exchanger for obvious reasons. Therefore, the purpose of the experiments was not to validate the internal flow, but to validate the temperature levels in the solid parts and the outflow temperature, i.e. the key parameters related to the functionality of the heat exchanger.

2.3. 3D printing

The heat exchanger examined in this study was 3D printed using aluminium (AlSi10Mg) with powder bed fusion (PBF) technique (Edgar and Tint, 2015). The heat exchanger was printed by the company FIT AG. PBF is one of the seven 3D printing techniques, and metal 3D prints are commonly done using it. In PBF, thin layer of metal or plastic powder is spread using a blade. Then laser, electron beam or thermal printhead selectively melts areas guided by information derived from a CAD file. Then new layer of powder is spread and the melting process is repeated, fusing the layers together. This is repeated layer by layer, until the whole part is printed. The printed heat exchanger was untreated without finishing work, such as milling. The piece was removed from the build platform by sawing. The part was constructed in $50 \mu\text{m}$ layers. The 3D printed object can be seen in Fig. 2c.

The benefits of 3D printing is that it allows manufacturing or nearly arbitrary shapes for an object as a single piece. In addition, the same CAD-model that is used for the 3D printing, is used to create the mesh

Table 1
The measures of the geometry.

Measure	L	D_i	D_o	h	W	l	w	t_f	t_w
Length/(mm)	260	10	14	9	104	200	3.6	1.5	2
Length/ D_i	26	1	1.4	0.9	10.4	20	0.36	0.15	0.2

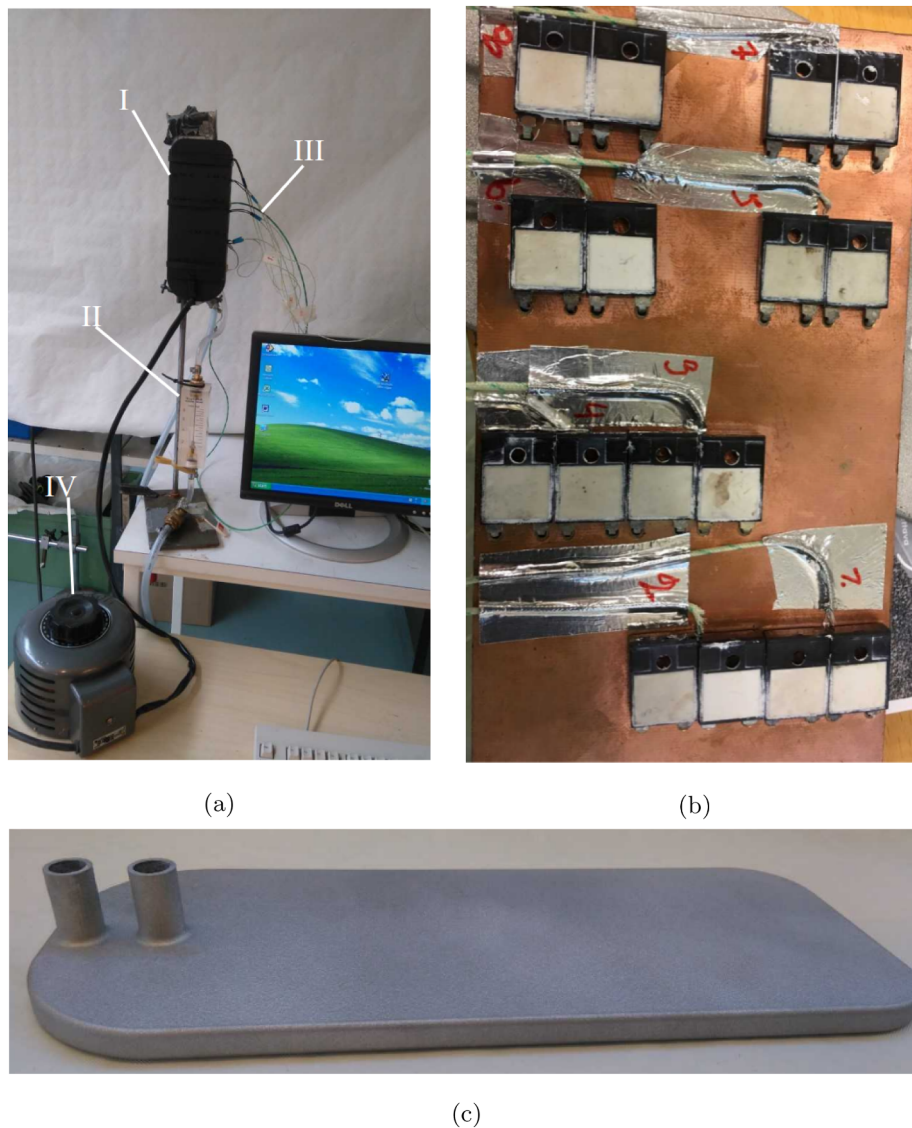


Fig. 2. (a): the experimental set-up used for the measurements. I: The heat exchanger, II: flow meter, III: thermocouple cabling, IV: power transformer. (b): The locations of the thermocouples used in the measurements. (c): The 3D printed heat exchanger.

for the simulations. This increases the similarities between the simulations and the experimental test case.

3. Computational set-up

3.1. The mesh and the initial conditions

The simulations were executed first with three different meshes to ensure that the quality of the mesh is not deteriorating the results. The selected mesh contains approximately 10.9 M cells in the fluid part and 3.2 M cells in the solid part of the domain. Due to the complexity of the geometry, it is challenging to create meshes with arbitrary cell size. The present mesh poses the highest spatial resolution that was still considered to be computationally feasible. Based on the numerical tests, the RANS models perform well with the coarser meshes. However, here the fine mesh is chosen for all turbulence modeling approaches. Further discussion on the mesh sensitivity for the present LES numerics is provided in Appendices A and B.

The cells are hexagonal and mostly isotropic. In the core parts, the cell sizes are 1 mm and 0.25 mm in the solid and fluid regions, respectively. Importantly the mesh is refined towards the solid–fluid interface for capturing the surface heat flux and the boundary layer

phenomena. In a single channel, with x-axis being in the main flow direction, the mesh contains approximately $N_y = 22$ and $N_z = 27$ cells in y and z directions, respectively, and approximately $N_x = 800$ cells in the x direction (see Fig. 1a). Such resolution is somewhat finer than used previously by Peltonen et al. (2019) in 3D heat transfer simulations (without CHT) in comparable Reynolds number conditions for LES in an air cooling heat exchanger. The number of cells in the inlet pipe along the diameter is approximately 48 with boundary layer refinement. The mesh was generated with the STAR-CCM+ software.

The analysis was based on a scenario with a total power of 400 W, corresponding a heat flux of $q'' = 83600 \text{ W/m}^2$ at the localized heat sources. A somewhat arbitrary limit for the maximum temperature for the electronic components to minimize degradation due to heating is $T_{max} = 65^\circ\text{C}$. The heated surface represents the temperature of the heat source. Hence, the heated surface should be maintained substantially below 65°C . The material properties of the cooling fluid and solid heat exchanger resemble those of water and aluminium at 27°C temperature, respectively. The thermophysical properties are listed in Table 2 and are assumed to be temperature independent in the present simulations. In the simulations, the flow rate is fixed to 2 L/min, which corresponds to mean inlet velocity of $U_{in} = 0.424 \text{ m/s}$ and Reynolds number of $Re \approx 4960$. Later on, in the last part of the results section, the

Table 2
The thermal properties used in the simulations.

Thermophysical property	Fluid	Solid
Density [ρ]	997 kg/m ³	2700 kg/m ³
Thermal conductivity [κ]	0.606 W/mK	209 W/mK
Specific heat capacity [c_p]	4180 J/kgK	900 J/kgK
Dynamic viscosity [μ]	$8.53 \cdot 10^{-4}$ Pa's	–
Prandtl number [Pr]	5.86	–

simulations are repeated with varying flow rate to address scalability of the system.

3.2. Governing equations

The fluid in the present study is modeled as incompressible. Both, the fluid and the solid domains are modeled with constant thermo-physical properties, as provided in Table 2. The present CHT simulations consist of solving the incompressible flow equations in the fluid domain while solving for convection–diffusion equation for temperature in the fluid and solid domains. In the fluid domain, the continuity equation can be written as

$$\nabla \cdot \mathbf{u} = 0 \quad (1)$$

where $\mathbf{u}$ is the fluid velocity which evolves according to the Navier–Stokes (NS) equation

$$\rho_f \left(\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot (\mathbf{u}\mathbf{u}) \right) = -\nabla p + \mu \nabla^2 \mathbf{u} \quad (2)$$

where p is the pressure and ρ and μ are the density and dynamic viscosity of the fluid (White and Corfield, 2006). The temperature is solved with the heat equation, which can be written as

$$\frac{\partial T}{\partial t} + \nabla \cdot (\mathbf{u}T) - \frac{\kappa_f}{\rho_f c_{p,f}} \nabla^2 T = 0 \quad (3)$$

and in the solid domain it reduces to

$$\nabla^2 T = \frac{\rho_s c_{p,s}}{\kappa_s} \frac{\partial T}{\partial t} \quad (4)$$

At the solid–fluid interface, the heat fluxes must satisfy the conservation of energy, i.e.,

$$\kappa_f \frac{\partial T_f}{\partial n} = \kappa_s \frac{\partial T_s}{\partial n} \quad (5)$$

and

$$T_f = T_s. \quad (6)$$

3.3. Numerical approach

3.3.1. Overview of the numerical approach

The continuous partial differential equations were discretized using the second order accurate finite volume method (FVM) with the open source C++ library OpenFOAM (Weller et al., 1998). Both, the RANS and LES simulations were carried out using a CHT solver `chtMultiRegionFoam`. The CHT method is based on calculating the conservation of mass, momentum and energy simultaneously in the fluid and solid regions using steady state (RANS) or transient (LES) assumptions. The CHT solver and the order of accuracy are further verified in Appendices A and B. For the pressure–velocity coupling, the Semi-Implicit Method for Pressure Linked Equations (SIMPLE) (Patankar and Spalding, 1983) algorithm was used in the steady state RANS simulations. SIMPLE is a two-step algorithm consisting of a prediction step and a correction step. The PIMPLE algorithm, which is a combination of SIMPLE and the Pressure Implicit with Splitting of

Operators (PISO) (Issa, 1986) algorithms, was used in the transient simulations. The main difference between PISO and SIMPLE is that PISO adds an extra correction step, resulting in an accelerated convergence rate of a single iteration, thus increasing its applicability for unsteady flows.

3.3.2. Turbulence models

Here, two RANS models and a Large-Eddy Simulation model were utilized. As mentioned, the RANS simulations were executed using the standard k - ε and k - ω SST models. The standard k - ε model is based on calculating the flow field with two equations, one for the turbulent kinetic energy, k

$$\rho \frac{\partial k}{\partial t} + \rho U_j \frac{\partial k}{\partial x_j} = \tau_{ij} \frac{\partial U_i}{\partial x_j} - \rho \varepsilon + \frac{\partial}{\partial x_j} \left[\left(\mu + \mu_T / \sigma_k \right) \frac{\partial k}{\partial x_j} \right] \quad (7)$$

and one for the dissipation of turbulent kinetic energy, ε

$$\rho \frac{\partial \varepsilon}{\partial t} + \rho U_j \frac{\partial \varepsilon}{\partial x_j} = C_{\varepsilon 1} \frac{\varepsilon}{k} \tau_{ij} \frac{\partial U_i}{\partial x_j} - C_{\varepsilon 2} \rho \frac{\varepsilon^2}{k} + \frac{\partial}{\partial x_j} \left[\left(\mu + \mu_T / \sigma_\varepsilon \right) \frac{\partial \varepsilon}{\partial x_j} \right] \quad (8)$$

where $\mu_T = \rho C_\mu k^2 / \varepsilon$ is eddy viscosity, U_i is average velocity component,

$$\tau_{ij} = \mu_T \left(\frac{\partial U_i}{\partial x_j} + \frac{\partial U_j}{\partial x_i} - \frac{2}{3} \frac{\partial U_k}{\partial x_k} \delta_{ij} \right) - \frac{2}{3} \rho k \delta_{ij} \quad (9)$$

is the Reynolds stress tensor, $C_{\varepsilon 1} = 1.44$, $C_{\varepsilon 2} = 1.92$, $C_\mu = 0.09$, $\sigma_k = 1.0$ and $\sigma_\varepsilon = 1.3$ (Shih et al., 1995; Wilcox, 1988; Menter, 1994).

The original k - ω model may be a better candidate for the viscous sublayer treatment. However, as a trade-off it may also be rather sensitive to the free stream value of ω (Menter, 1992). Similar to the k - ε model, the k - ω model is a two-equation model, where ω corresponds to the specific dissipation rate of turbulent kinetic energy. The k - ω SST model, developed by Menter (1994), performs as the original k - ω model near the boundaries, and switches into the standard k - ε in the free stream regions. The transport equations for k and ω are

$$\rho \frac{\partial k}{\partial t} + \rho U_j \frac{\partial k}{\partial x_j} = \tau_{ij} \frac{\partial U_i}{\partial x_j} - \beta^* \rho \omega k + \frac{\partial}{\partial x_j} \left[\left(\mu + \mu_T \sigma_k \right) \frac{\partial k}{\partial x_j} \right] \quad (10)$$

and

$$\rho \frac{\partial \omega}{\partial t} + \rho U_j \frac{\partial \omega}{\partial x_j} = \frac{\gamma}{\nu_T} \tau_{ij} \frac{\partial U_i}{\partial x_j} - \beta \rho \omega^2 + \frac{\partial}{\partial x_j} \left[\left(\mu + \mu_T \sigma_\omega \right) \frac{\partial \omega}{\partial x_j} \right] + 2 \left(1 - F_1 \right) \rho \sigma_{\omega 2} \frac{1}{\omega} \frac{\partial k}{\partial x_j} \frac{\partial \omega}{\partial x_j} \quad (11)$$

where the constants are $\sigma_{k1} = 0.85$, $\sigma_{k2} = 1$, $\sigma_{\omega 1} = 0.5$, $\sigma_{\omega 2} = 0.856$, $\beta_1 = 0.075$, $\beta_2 = 0.0828$, $\beta^* = 0.09$, $\gamma = \beta / \beta^* - \sigma_{\omega 1} \kappa^2 / \sqrt{\beta^*}$, $\kappa = 0.41$, ν_T is the kinematic eddy viscosity and F_1 is a function that is one in the near wall region and zero further away from the boundary. The constants are calculated according to F_1 by

$$\phi = F_1 \phi_1 + (1 - F_1) \phi_2 \quad (12)$$

where ϕ is one of the constants in Eqs. (10) and (11) (Menter, 1994). The k - ω SST model performs well for low Reynolds number flows since it limits the production of turbulence by considering the transport of the turbulent shear stress. Hence, it is chosen as the base method for the present study. With both RANS models, the turbulent heat fluxes are modeled with a constant turbulent Prandtl number of 0.85 being the default value in OpenFOAM.

In OpenFOAM, the wall treatment in the RANS models depends on the value of y^+ in the near-wall cells. The wall functions are activated only if $y^+ > 11$ (Liu, 2016). Here, a standard wall function for the turbulent viscosity is used, which switches to zero when $y^+ < 11$. Effectively, this corresponds to wall damping. In the present study, $y^+ < 1$ and hence, the turbulent viscosity is set to zero at all boundaries. In general, we note that the Reynolds number in the channels is relatively low $Re \approx 925$.

Additionally, the Large-Eddy Simulation (LES) approach was utilized to simulate transient flow. In LES the grid scale turbulence is directly resolved while the subgrid scale turbulence is modeled. Here, similar to our previous study (Peltonen et al., 2019), the implicit LES (ILES) was adopted. In the present ILES the numerical dissipation is controlled by parameter $\gamma \in [0, 1]$ for the convective terms. The introduced numerical diffusion of the γ -scheme by Jasak et al. (1999) functions as a stand-alone turbulence model. The maximum numerical diffusion is obtained at $\gamma = 1$ while $\gamma = 0$ corresponds to a non-diffusive linear scheme. In the present ILES, similar subgrid-scale dissipation is assumed for momentum and heat. In this study, the value $\gamma = 0.2$ is used.

The present ILES has been used previously in various studies by the authors in various applications. In heat transfer context, the method was used recently by Peltonen et al. (2019). In laminar and turbulent pipe flow context, the method was tested and validated by Laurila et al. (2019). Further demonstration on the functionality of the ILES model is provided in Appendices A and B. Additionally, sensitivity assessment of numerical results on the value of γ as well as grid resolution have been provided by Peltonen et al. (2019) for a pin-fin configuration.

4. Results and discussion

4.1. Overview on flow characteristics

Certain key flow characteristics in the heat exchanger in addition to separated flow regions are shown in Fig. 3a and b. The flow phenomena are denoted with roman numerals (i–ix) while the flow regions are marked with letters (A–E). The instantaneous streamlines in Fig. 3a are colored with the temperature of the fluid. The flow exhibits versatile

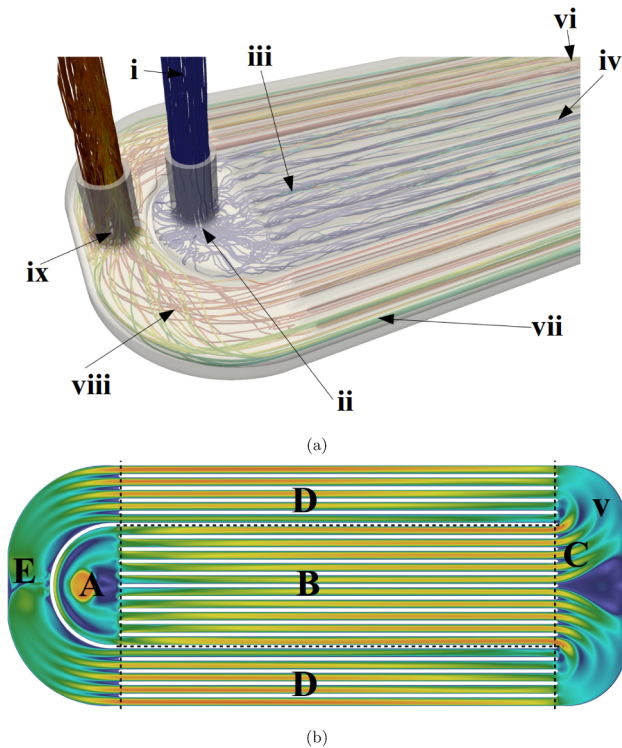


Fig. 3. (a): Streamlines of an instantaneous velocity field colored by the temperature. (i): Turbulent inlet (ii): Impinging jet (iii): Decaying turbulence (iv): Laminarized flow (v): Transition to turbulence (in panel b) (vi): Decaying turbulence (vii): Re-laminarized flow (viii): Re-transition to turbulence (ix): Swirl (b): The different regions for flow analysis separated by a dashed line. (A): Inlet region (B): Inner channels (C): Reversal region (D): Outer channels (E): Outlet region. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

physics with successive regions of turbulence, laminarized flow, transition to turbulence, re-laminarization and re-transition. Turbulent inflow (i) enters the domain as a jet ($Re \approx 4960$) that impinges at the opposite wall (ii). The impinging jet spreads along the base plate and enters the inner coolant channels. In the channels, the initially turbulent flow (iii) laminarizes (iv) due to the low Reynolds number of the channels ($Re \approx 925$). Decaying turbulent flow has been reported to exhibit higher rate of heat transfer in the entrance region compared to a developing laminar flow (Cornejo et al., 2019). At the end of the inner coolant channels, the laminar flow undergoes transition to turbulence (v) (Fig. 3b). In the outer coolant channels, the turbulent flow (vi) re-laminarizes (vii), and undergoes re-transition to turbulence (viii). The instantaneous flow may occasionally swirl in the outlet pipe (ix). However, the mean flow does not exhibit clear swirl due to the flow symmetry.

The domain of the heat exchanger is divided into five different regions, as seen in Fig. 3b. The fluid enters the domain in the inlet region (A). From the inlet region, the fluid enters the inner channels of the heat exchanger (B), where the convective heat transfer takes place. The channels are numbered so that the inner channels are channels number 6–15, and the outer channels are number 1–5 and 16–20. The fluid takes a U-turn in the reversal region (C) from the inner channels towards the outer channels (D). Finally, the fluid exits the domain from the outlet region (E). Due to the flow symmetry and absence of heat sources, the flow characteristics in the outlet region are assumed to be insignificant for the overall thermal performance of the heat exchanger.

After the inlet region, the flow is confined into narrow channels where the Reynolds number varies within the range of $Re = 925 \pm 100$ depending on the channel number. The laminarization along the channels was noted in the velocity field in Fig. 3a. In Fig. 4, turbulence intensity is plotted along a single channel (channel 10). The figure inset shows a cut-plane of the velocity field inside the channel. The laminarization is clearly noted from the decay of the turbulence as well as the instantaneous flow visualization in the LES. The laminarization is not predicted by the RANS models. It is noted that the $k-\omega$ SST model results compare relatively well with the present ILES inside the channels. Such favorable correspondence may be partially related to two counter-acting effects: underestimated turbulence at the entrance of the channel and over-estimated turbulence at the outlet of the channel. In the present study, heat transfer occurs in the turbulent, transitional and laminar parts of the channels. The effect of the laminarization on the heat transfer performance is discussed further in chapter 5.

4.2. Flow structures

4.2.1. Flow in the inlet region

In the present heat exchanger, the inlet and outlet are perpendicular

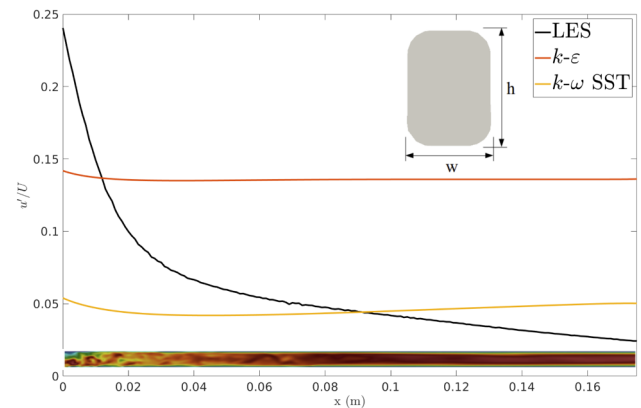


Fig. 4. Turbulence intensity along a channel. The figure contains also the cross-section of the channel and a visualization of the instantaneous velocity field. The figure reveals laminarization with the LES.

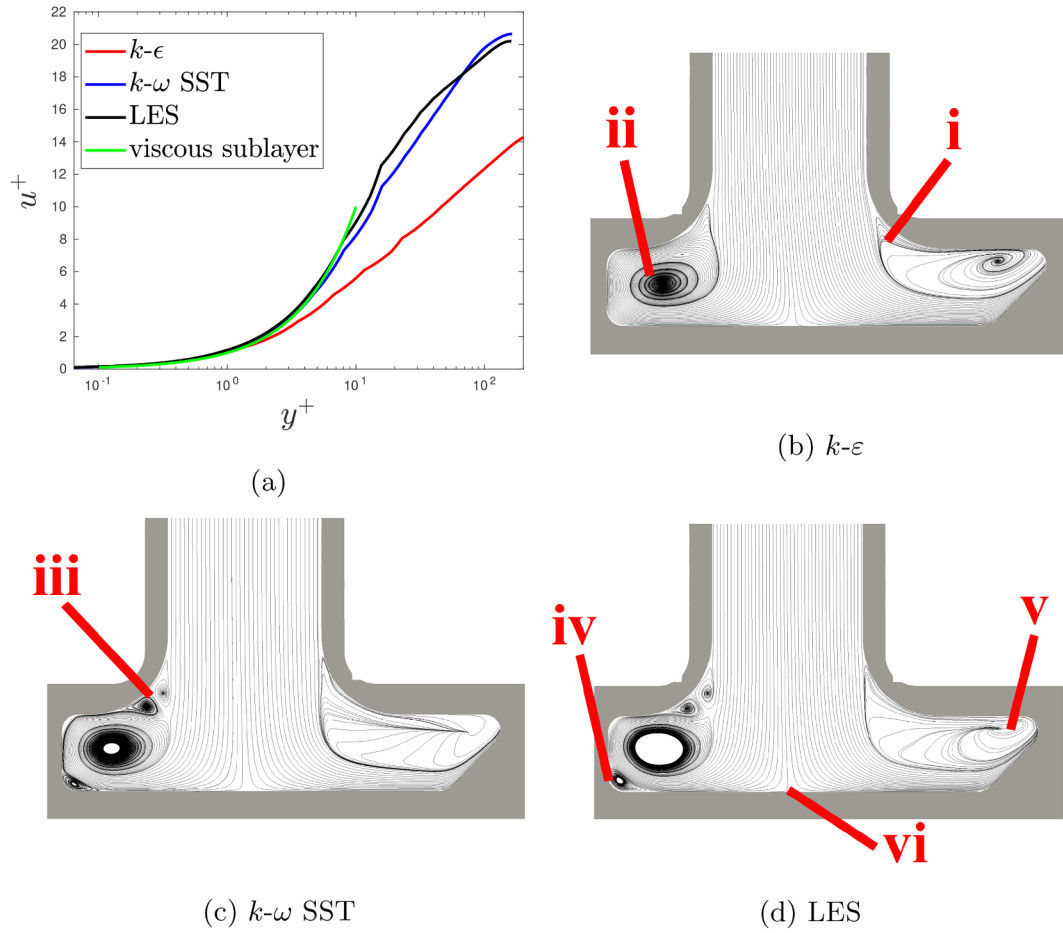


Fig. 5. (a): Velocity distribution in the inlet pipe in all simulations. (b): Streamlines in the inlet area with $k-\epsilon$. (c): Streamlines in the inlet area with $k-\omega$ SST model. (d): Streamlines in the inlet area with LES. (i): Flow separation. (ii): Recirculation vortex. (iii): Recirculation vortices. (iv): Corner vortex. (v): Mixing vortex. (vi): Stagnation point. The figure reveals a more diffusive solution with the $k-\epsilon$ model and more small scale details with LES and $k-\omega$ SST.

to the cooling plate for achieving higher flow uniformity in the coolant channels. For realistic boundary condition modeling, fully developed turbulent flow in the inlet pipe is simulated. The mean velocity profiles in the inlet pipe can be seen in Fig. 5a. The turbulent velocity distributions are non-dimensionalized by the friction velocity, $u_\tau = \sqrt{\tau_w/\rho}$, where τ_w is the wall shear stress, and plotted against non-dimensional wall distance, y^+ . The friction velocity is different for each simulation. In general, it is important to properly model the velocity and temperature boundary layer characteristics. It is noted, that the velocity profile simulated with $k-\epsilon$ model differs from the other models the most.

As mentioned, the present flow system poses relatively versatile fluid dynamical features. Such aspects advocate the usage of turbulence models that are capable of capturing the related flow dynamics. The key flow structures in the inlet region for all three turbulence models can be seen in Fig. 5b, c and d. The flow can be divided into primary flow structures, such as region of separation (i) and recirculation vortex (ii). Additionally, secondary flow structures (iii) and (iv), can be noted in the vicinity of the primary structures for $k-\omega$ SST and LES, while the standard $k-\epsilon$ model does not capture such details. Mixing of the fluid (v) before the channels can be seen with all the models. The average value of y^+ at the domain walls with all the models was well below 0.5, while the maximum value was $y^+ \approx 2$ at the location of the jet impingement.

4.2.2. Flow in the reversal region

In the reversal region, the flow exits the inner channels, takes a U-turn, and reverses direction towards the outer channels. The flow distribution in the outer channels depends on the flow characteristics in the reversal region. Thereby, it is of interest to understand the flow

structures in this region (see objective 2).

The streamlines based on planar field data in the reversal region are shown in Fig. 6. The streamlines show similarities between the models, revealing consistent primary flow structures (i). In addition, similar secondary flow structures, such as fin tip vortices (ii) and flow separation (iii) can be observed. Secondary recirculation (iv) in the vicinity of the primary flow structures can be seen with $k-\omega$ SST and LES. However, such details are not captured by the standard $k-\epsilon$ model.

4.2.3. Flow in the channels

The coolant channels in the domain are divided into inner channels, and outer channels. The flows in the inner and outer channels are in opposite directions. As mentioned, the outer channels are referred with numbers 1–5, and 16–20 while the inner channels are referred with numbers 6–15. Based on the conservation of energy, the flow rate in each coolant channel affects the rate of heat transfer. In addition, the flow properties are known to affect the local heat transfer. Hence, the uniformity of the flow rate in the channels is a desired feature. In the following, the flow distribution in the channels is discussed with linkage to the previously observed flow features.

In this paper, the flow velocity in the channels refers to the non-dimensional absolute velocity magnitude. The velocity is non-dimensionalized by the average velocity in the channels, i.e., the sum of the non-dimensional velocities equals unity

$$\sum_i u_i/U = 1$$

The velocity distribution is shown in Fig. 7. The Fig. 7d includes an

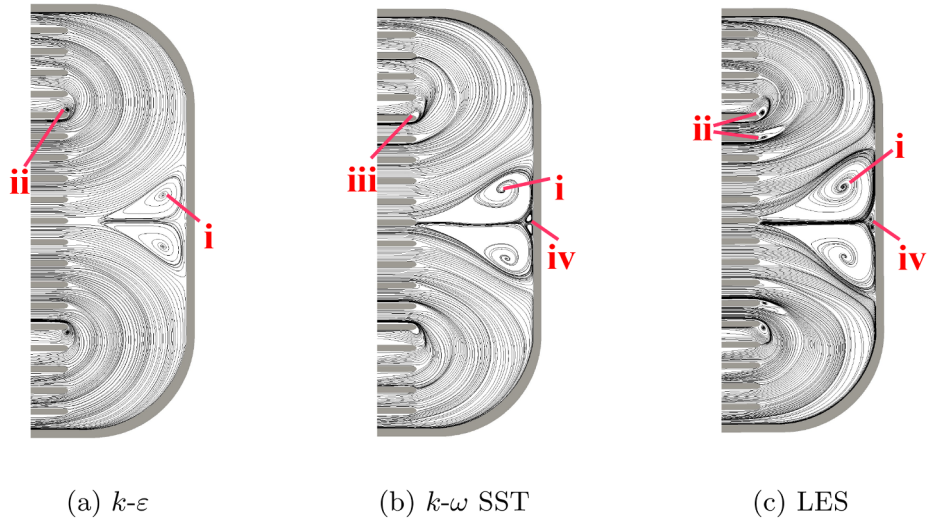


Fig. 6. Streamlines in the reversal region. (i): Primary flow structure. (ii): Fin tip vortices. (iii): Flow separation. (iv): Secondary flow recirculation. The figure shows more small scale details in the LES field and more diffusive solution with $k-\varepsilon$.

average over each model including the deviation between the highest and the lowest velocities. In general, the lowest velocities are observed in channels 5 and 16, while the highest velocities are in channels 1, 6, 15 and 20. The difference between the highest and the lowest velocities

is the smallest with $k-\varepsilon$ and the greatest with LES.

From the presented analysis, it is clear that there is a linkage between the small scale flow structures and the overall flow uniformity in the coolant channels. It was observed that the scale of the fin tip

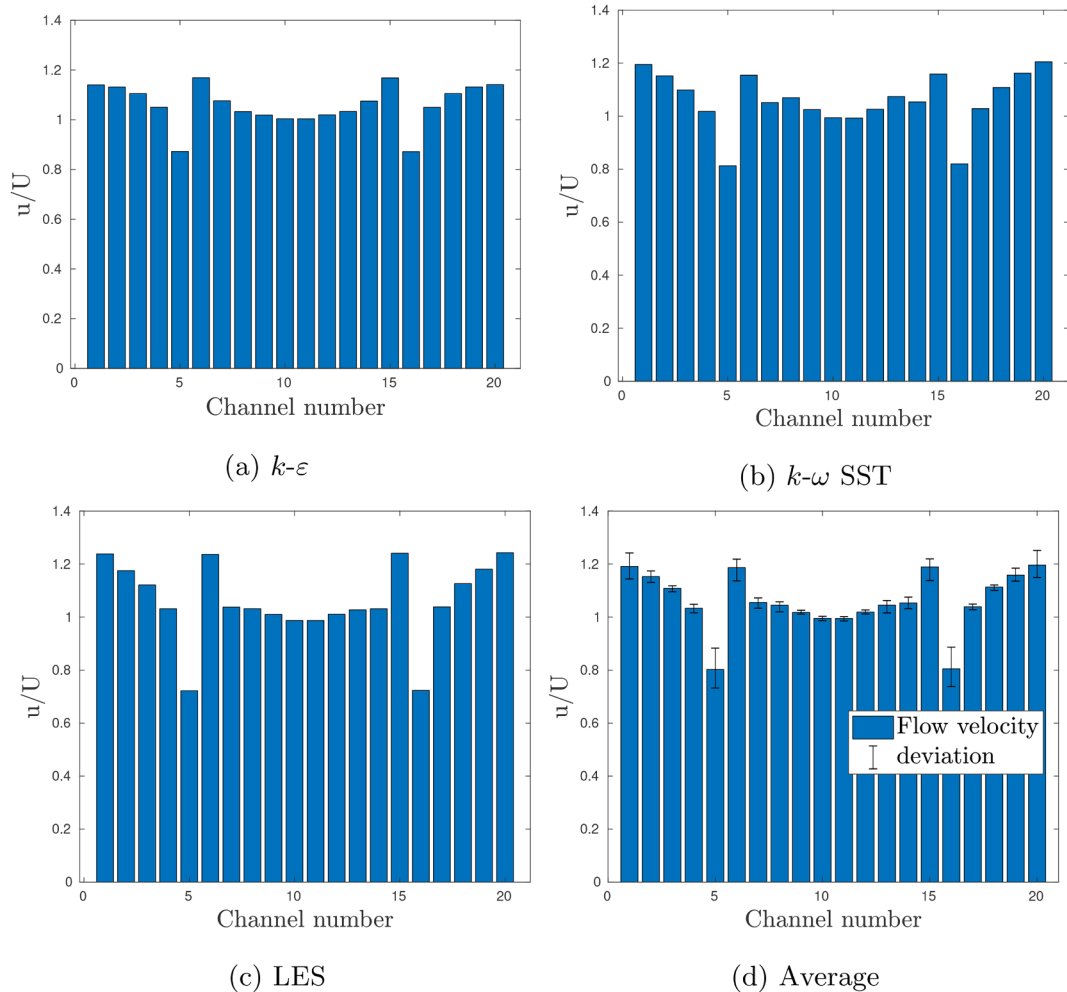


Fig. 7. The relative flow distribution in the channels. (a): $k-\varepsilon$ (b): $k-\omega$ SST (c): LES (d): Average value over all the simulations with the deviation showing the maximum and minimum values. The flow uniformity is overestimated with the standard $k-\varepsilon$.

vortices and primary flow structures in the reversal region is the smallest with the standard $k-\epsilon$ model and the largest with LES. This observation correlates directly with the velocity distribution in the outer channels (see Fig. 7). Flow separation at the location of the small fin tip vortices (see Fig. 6 ii–iii) distributes the fluid momentum outwards from channels 5 and 16, resulting in lower flow rates at these channels. The larger primary flow structures (see Fig. 6 i) increase the momentum of the fluid towards the wall of the heat exchanger, resulting in larger flow rates in channels 1 and 20. In addition, it was observed that some small scale secondary flow structures in the inlet region (see Fig. 5 iii–iv) were not captured by the standard $k-\epsilon$ model. This correlates with the flow uniformity in the inner channels. The corner vortices increase the momentum of the fluid near the boundaries, resulting in higher flow rates in channels 6 and 15. It is clear that the small and large scale flow structures are significant for the overall flow uniformity in the coolant channels.

4.2.4. Pressure drop

With relevance to the overall flow distribution and pumping losses, the pressure drop over the heat exchanger is calculated from the pressure difference between the inlet and the outlet. The corresponding pumping power is calculated from

$$P_{\text{pump}} = Q\Delta p \quad (13)$$

where Q is the volumetric flow rate and Δp is the pressure difference. The pressure losses with the corresponding pumping power for each model are listed in Table 3. The results reveal that the pumping power compared to the heating power (400 W) is insignificant to the thermal efficiency of the heat exchanger.

4.2.5. A remark on RANS models

As noted previously, there are differences between the flow characteristics of the standard $k-\epsilon$ model and the $k-\omega$ SST model. Especially, the small scale characteristics in the near-wall regions are noted to differ. One key difference between the models is the limitation of the production of turbulence in the boundary layer in the $k-\omega$ SST model (Menter, 1994). Since the case at hand is mostly dominated by boundary layer effects, it is presumable that the standard $k-\epsilon$ model does not result in an accurate prediction of the flow characteristics. In addition, the standard $k-\epsilon$ model overestimates the turbulent length scale in the near-wall region in the presence of adverse pressure gradient (Wilcox, 1988). Fig. 8 shows the eddy viscosity in a cut-plane near the inlet region for $k-\epsilon$ and $k-\omega$ SST models. It is noted, that the turbulent viscosity is larger with $k-\epsilon$ model, especially in the vicinity of the jet impingement. As noted in Fig. 5, the impinging jet creates recirculation of the flow. The recirculation occurs due to the presence of adverse pressure gradient, resulting in the overestimation of the turbulence with the standard $k-\epsilon$ model. The higher values in the turbulent viscosity result in a more dissipative solution. The implicit LES is calculated without turbulent viscosity.

4.3. Temperature analysis

At steady state, the temperature levels of the heat exchanger are of paramount importance for the purpose of cooling. The rate of cooling depends on the flow rate of the cooling water and the temperature

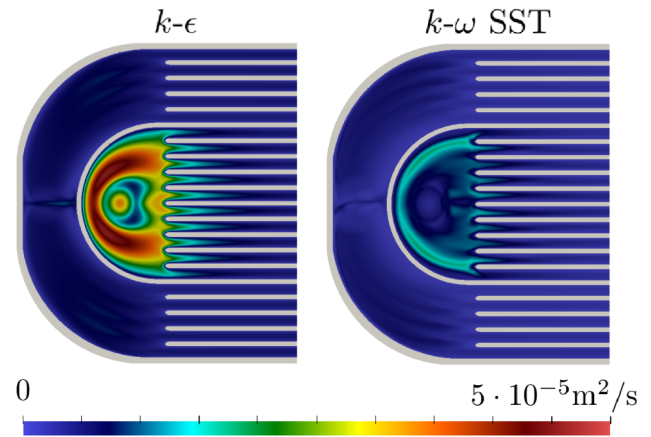


Fig. 8. Turbulent viscosity in a cut-plane from the centre of the domain with $k-\epsilon$ and $k-\omega$ SST models. The figure reveals significantly higher turbulent viscosity with the standard $k-\epsilon$ model.

difference between the fluid and the solid. Previously, we noted that the flow distribution is relatively uniform in the channels within 15–25% accuracy. The most significant aspects for the temperature analysis are the temperature difference between the fluid and the solid, and the hot-spot temperatures at the heated surface. The following analysis focuses on model-to-model variations in temperature prediction.

4.3.1. Solid–fluid interface

The temperature field of a cut-plane at the centre of the heat exchanger can be seen in Fig. 9. The figure shows higher temperatures in the solid and lower in the fluid at the locations of the heat sources. In the outlet region, the bulk temperatures (i) of the fluid have reached the highest levels. The figure reveals that the variance in the temperatures in this region is the smallest with $k-\epsilon$ model and the greatest with LES. This is expected due to the higher dissipation of the solution with the standard $k-\epsilon$ model. The temperature levels in the solid part between $k-\omega$ SST and LES are noted to be relatively similar. However, the discrepancy between the peak solid temperatures between the standard $k-\epsilon$ model and the other models is approximately 5 °C. The temperature variance in the reversal region (ii) is higher with LES compared to $k-\omega$ SST, however, the hot-spot temperatures (iii) are noted to be almost identical. With all the models, the temperature in the solid domain is calculated with a similar procedure, i.e., the differences in the solid temperature levels between each simulation is solely due to the temperature levels of the solid–fluid interface. The dissipation of the solution with the standard $k-\epsilon$ model produces additional mixing of the temperatures in the coolant channels. Hence, the boundary layer temperatures of the fluid are underestimated, thus decreasing the solid temperatures. This indicates the importance of using a CHT approach instead of e.g. a constant temperature or constant flux boundary condition for the fluid side. In addition, it indicates the importance of capturing the small scale flow structures for a realistic modeling of heat transfer.

Fig. 10 shows line probed temperature profiles in normal direction of the cooling plate at locations (i) and (ii). The locations of the lines are based on the highest and the lowest heat fluxes in the heat exchanger. The first graph (i) corresponds to the highest heat flux. Fig. 10b shows a steep temperature gradient near the boundary. The temperature profiles between $k-\omega$ SST and LES have no significant differences, and the temperature profile near the boundaries is noted to be almost identical. The temperature gradient near the boundaries with $k-\epsilon$ is noted to be less steep. The second graph (ii) corresponds to the lowest heat flux. In this location, the heat flux reaches negative values, i.e., heat is transferring from the fluid to the solid. This can be seen from the temperature values near the boundaries. The temperature of the fluid is higher

Table 3

The pressure difference between the inlet and the outlet of the heat exchanger with the corresponding pumping power with the different models.

Model	Δp (Pa)	P_{pump} (mW)
$k-\epsilon$	443	14.6
$k-\omega$ SST	326	10.8
LES	340	11.2

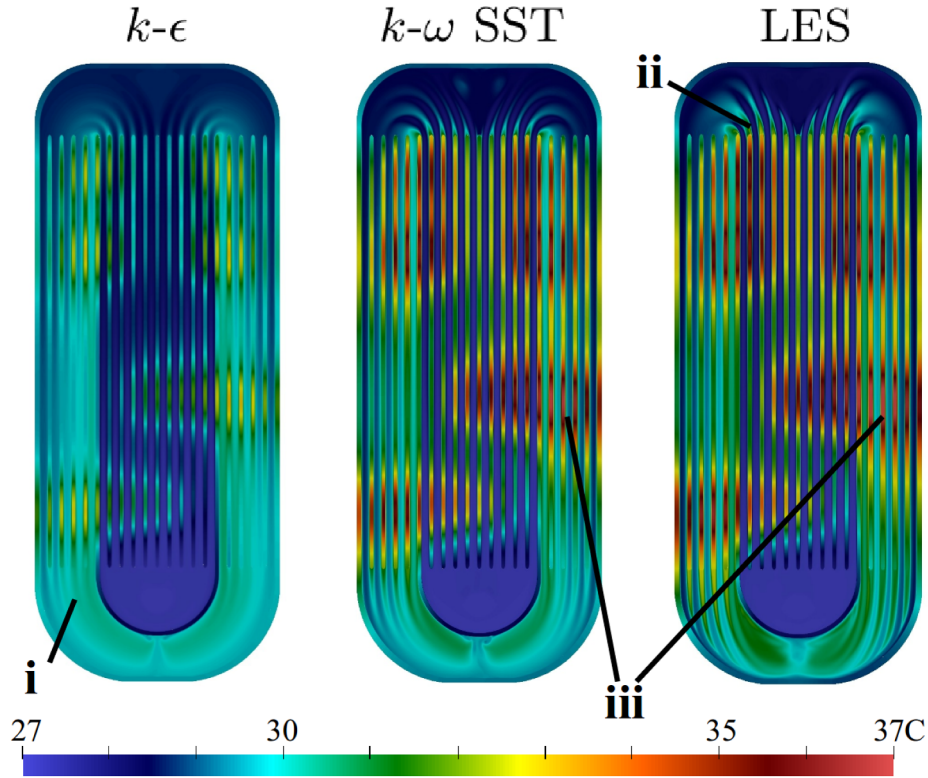


Fig. 9. The temperature field of a cut-plane at the centre of the domain including both, the fluid and solid regions. (i): Uniform temperature levels in the outlet area with $k-\epsilon$ model. (ii): Mixing of temperatures in the reversal region. The steepest temperature gradients with LES. (iii): Highest temperatures in the solid in the downstream. The figure reveals more similarities in the temperature levels between $k-\omega$ SST and LES compared to $k-\epsilon$.

than the solid in the boundary layer, and hence, the heat must transfer from the fluid to the solid. The flow is laminar at this location, and hence, the temperatures in the boundary layer mix slowly by diffusion, and the higher temperatures in the boundary layer convect persistently downstream of the heat sources. The temperature profile of the $k-\epsilon$ model deviates from the other profiles. The most significant difference is that the temperature profile shape is completely different from the other models, since it does not exhibit the temperature peaks in the boundary layers. This indicates falsely predicted thermal diffusion by the $k-\epsilon$ model. Additionally, a subtle difference in the temperature levels between $k-\omega$ SST and LES can be noted, however, the shape of the profiles are highly similar.

Fig. 10d and e show the ratio κ_T/κ_f i.e. the turbulent thermal conductivity divided by the fluid thermal conductivity. The turbulent thermal diffusivity is calculated based on the turbulent Prandtl number and the turbulent dynamic viscosity

$$Pr_T = \mu_T/\alpha_T \quad (14)$$

where $\alpha_T = \kappa/c_p$ is the turbulent thermal diffusivity. The turbulent Prandtl number is commonly defined based on the turbulent kinematic viscosity ν_T , however, with the present solver's definition for α_T , turbulent dynamic viscosity is used. The turbulent thermal conductivity is calculated from the turbulent thermal diffusivity. The effective heat flux at the boundary is a sum of the conductive heat flux and the turbulent heat flux

$$q''_{eff} = \left(\kappa_f + \kappa_T \right) \frac{\partial T}{\partial n}. \quad (15)$$

The figures show overestimated turbulence with $k-\epsilon$ model compared to $k-\omega$ SST model. Since the effective heat flux is determined by the energy balance, the overestimation of the turbulence decreases the temperature gradient. With $k-\omega$ SST model the turbulent thermal conductivity near the boundaries is very low, indicating that the heat

transfer near the boundaries is resolved, consistent with Fig. 10b. The values for the turbulent thermal conductivity are obtained from the cell points and not the cell faces. The value of the turbulent thermal conductivity at the boundary face is zero.

4.3.2. Surface temperature distribution

The temperature distributions on the heated surfaces are shown in Fig. 11. As can be noted, the surface temperatures are significantly lower with the $k-\epsilon$ model, yet the $k-\omega$ SST and LES are noted to be similar, consistent with the previous analysis. The lower surface temperature levels with the $k-\epsilon$ model can be explained by the underestimated temperature gradients. In the figure, the borders of the individual channels can be hardly seen (i). Two different types of heat conduction in the solid are observed based on the flow type. Heat conducts in the solid to the stream-wise direction from the hot-spots (ii) in the laminar regime where diffusion is dominating in both, the fluid and solid, regions. A similar effect cannot be seen in the turbulent regime (iii) where turbulence transfers heat efficiently from the boundary layer to the bulk flow. The bulk temperature of the water is increasing due to heating, and hence, warmer temperatures in the solid can be seen in the downstream areas (iv).

The maximum hot-spot temperatures, along with the average temperature of the heated surface are listed in Table 4. The maximum surface temperatures reach values of $\sim 40^\circ\text{C}$, which is much lower than the temperature limit of 65°C .

The solution for the temperature distribution reveals the importance of using CHT analysis instead of a constant temperature or constant heat flux boundary conditions. Firstly, it was noticed, that in some downstream regions, the heat flux was from the fluid to the solid. This physical detail could not have been resolved correctly without considering the solid conduction. Secondly, the model-to-model peak boundary temperature variation was noted to be significant ($\approx 5^\circ\text{C}$). Finally, the temperatures at the heated surface of the solid are of great

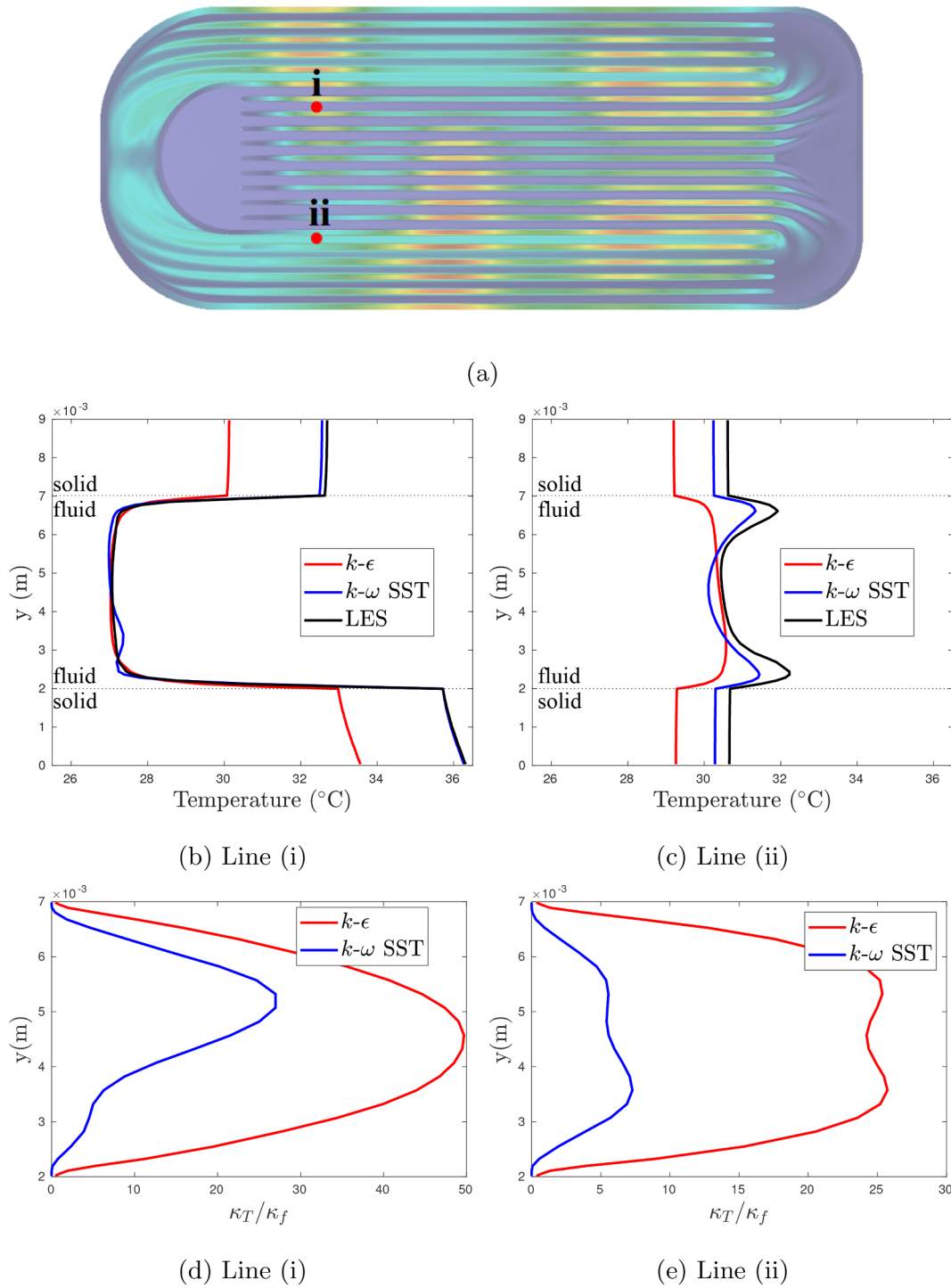


Fig. 10. Temperature ((b) and (c)) and the ratio of the turbulent thermal conductivity to the material thermal conductivity κ_T/κ_f ((d) and (e)) through the domain in the normal direction at two specific locations (i) and (ii). The figure shows overestimated turbulence and underestimated temperature gradients with the standard $k-\epsilon$ model.

importance in the functionality of the heat exchanger. These temperatures could not be solved simultaneously without CHT approach. It is noted, that there is no significant local temperature differences seen in the solid domain due to the variation of the velocity in the channels.

In conclusion of the provided numerical temperature analysis, the liquid cooling heat exchanger performs well for the purpose it is designed for. The applicability of CFD as a design tool depends heavily on the simulation assumptions. In this case, dimensioning the heat exchanger based on the results with the standard $k-\epsilon$ model would lead to

higher surface temperatures in the physical heat exchanger than predicted based on the simulations.

4.4. Mass flow rate assessment

The mass flow rate sensitivity analysis is performed with the $k-\omega$ SST model. This is justified since the comparison between that model and LES revealed no significant difference in the temperature levels compared to the additional computational effort needed with LES. The mass

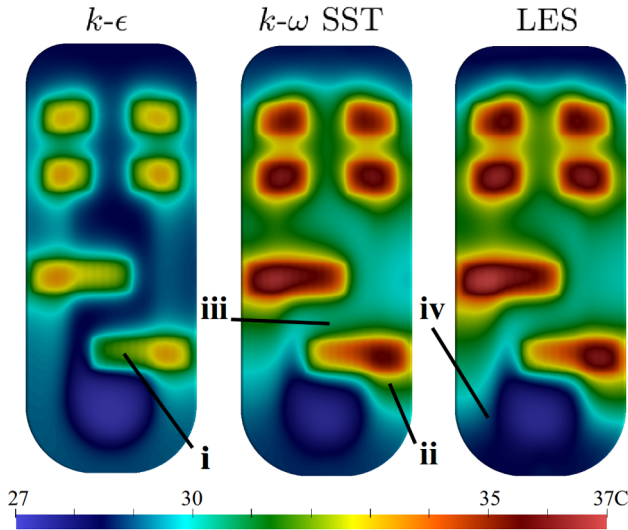


Fig. 11. Surface temperature. (a) $k-\epsilon$ (b) $k-\omega$ SST (c) LES. (i): Subtle temperature variance at the locations of the fluid channels, (ii): Conduction of the heat from the hot-spots in the stream-wise direction in laminarized areas, (iii): Little conduction of the heat from the hot-spots to the stream-wise direction of the fluid in the turbulent areas, (iv): Warmer temperatures in the solid due to the fluid warming the solid. The figure shows that $k-\epsilon$ underestimates the temperature levels.

Table 4

The maximum temperature levels and the average hot-spot temperatures in the simulations.

Model	$T_{s,max}/(^{\circ}\text{C})$	$T_s/(^{\circ}\text{C})$
$k-\epsilon$	35.4	33.4
$k-\omega$ SST	39.3	36.5
LES	39.9	36.8

flow rate assessment is performed in order to ensure that the velocity distribution in the channels does not depend on the flow rate.

The velocity distribution over the flow rates of 2, 3, 4, 5 and 6 L/min are shown in Fig. 12. In terms of inlet Reynolds numbers, this corresponds to the range of $4960 \leq Re \leq 14880$. The figure is very similar to Fig. 7d. However, in Fig. 12 the average is calculated over different flow rates for a single turbulence model, whereas in Fig. 7d the average is over the different turbulence models with a single flow rate. The figure also includes the difference between the maximum and minimum velocities in each channel between the different flow rates. The figure reveals no significant differences in the velocity uniformity with the change in the mass flow rate. This observation indicates that the velocity uniformity does not depend significantly on the inlet Reynolds number. This is important for achieving a smooth cooling process, since the cooling efficiency of the heat exchanger can be controlled directly by altering the flow rate and the inlet water temperature. In addition, the numerically demonstrated flow uniformity insensitivity on the inlet Reynolds number indicates scalability of the heat exchanger in the considered range of flow conditions.

4.5. Comparison between experiments and simulations

The experiments were carried out with flow rates of 2, 3, 4, 5 and 6 L/min, and a power 400 W. The experiments were performed to confirm the temperature levels obtained in the simulations. In Fig. 13, the thermal images from the experiments are compared with the simulated surface temperatures. The thermal images are from the bottom side of the test circuit while the simulations are from the heated surface of the heat exchanger. The electronic circuit is connected to the heat

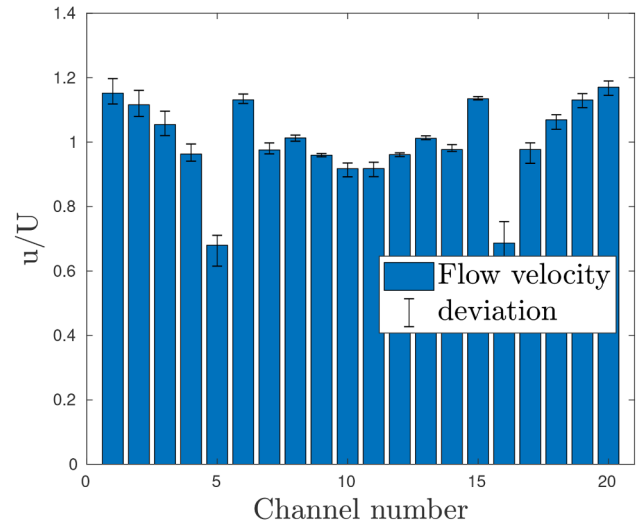


Fig. 12. Average non-dimensional absolute velocity for $k-\omega$ SST model over varying flow rate. The error bar indicates low flow rate sensitivity.

exchanger with thermal paste via the electronic resistors (iii). Hence, the circuit board, or the conducting connector parts of the circuit (i), are not being directly cooled. The surface of the heat exchanger between the hot-spots (iv) is at a lower temperature level compared to the circuit board (iii). One should consider that the experiments are not directly comparable to the simulations due to the difficulties of exact measurements.

The temperature levels of the resistors measured at the locations seen in Fig. 2b are listed in Table 5. The temperature levels reveal that the heat exchanger is able to provide efficient cooling. Comparing the values in Table 5 to Table 4 reveals similarities between the measured and simulated temperature levels.

4.6. Discussion

In Fig. 14a the temperature levels from the simulations and experiments are presented. The figure includes the average temperature of the heated aluminium heat exchanger surfaces from the simulations, the average temperature of all measured electric resistor temperatures, and an estimated corrected experimental temperature value for the heated surface temperature. The corrected value represents the temperature at the heated aluminium heat exchanger surface below the electric resistor and the thermal paste. The correction of the surface temperature is based on the thermal conductivity (8.5 W/mK), estimated thickness (0.1 mm) and contact area (2 cm²) of the thermal paste. Based on the Fourier's law, the temperature correction can be estimated to be 1.5 K. The figure also includes the simulated and experimental bulk outlet temperatures, and the corresponding analytic counterpart. The temperature levels are referenced to the inlet temperature. The figure reveals good agreement in the temperature levels between the simulations and the experiments.

The velocity uniformity was noted to be mostly independent of the inlet Reynolds number. Hence, the cooling power can be controlled directly with the inlet temperature and the mass flow rate. Such observation motivates the evaluation of the maximum inlet temperatures that would still maintain surface temperatures below 65 °C. In Fig. 14b the maximum inlet temperature is plotted against the flow rate indicating that the system would still be operational at inlet temperatures up to ~55 °C. The serial configuration of the heat exchangers represents a case where a total of four circuits are cooled using the outlet water of the previous heat exchanger as the inlet water for the next in the series. Since in a parallel configuration, the total number of heat exchangers does not affect the inlet water temperatures, the analysis of the parallel

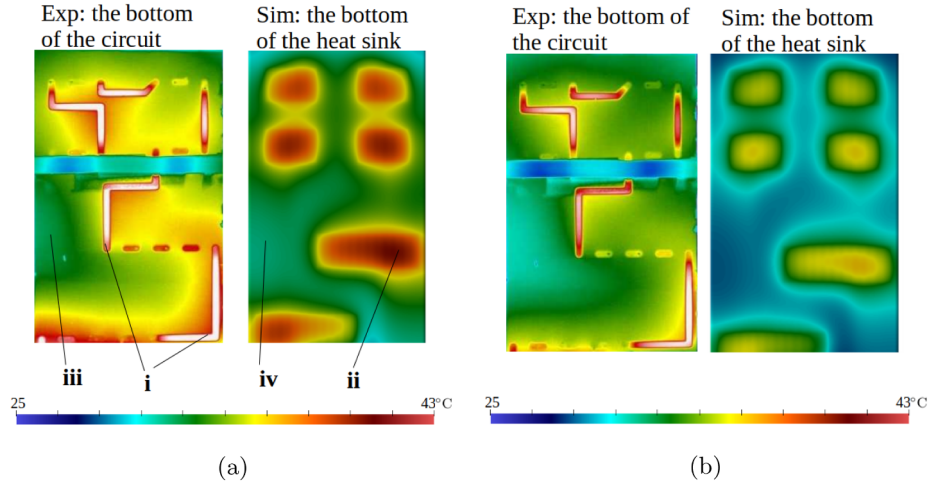


Fig. 13. Thermal imaged taken by an infrared camera compared to the simulated surface temperatures. (a): 2 L/min flow rate. (b): 4 L/min flow rate. (i): Conductors. (ii): Hot-spots. (iii): Surface of the circuit. (iv): Surface of the aluminium heat exchanger. The figure reveals similar hot-spot temperatures.

Table 5

The temperature levels measured with the thermocouples, locations of which are shown in Fig. 2b.

$Q/(L/min)$	T_{in}	T_{out}	T_1	T_2	T_3	T_4	T_5	T_6	T_7	T_8
2	25.5	28.3	37.1	37.2	37.0	38.5	38.7	37.2	35.4	34.2
3	26.8	28.8	36.9	37.3	37.3	38.9	38.5	37.8	35.3	34.8
4	27.0	28.5	35.7	36.2	36.0	37.4	37.2	36.3	34.0	33.5
5	26.1	27.1	34.7	35.3	34.7	35.9	35.9	34.9	33.0	32.2
6	26.7	27.7	35.1	35.7	34.6	35.8	36.0	34.8	33.3	32.3

configuration is based on one heat exchanger in series.

Based on the numerical results, turbulence model can have a relatively large impact on CFD-CHT results of the considered heat exchangers. In particular, the results demonstrate the well-known aspect that challenging flow characteristics near the walls and transition/laminarization phenomena need to be adequately captured. These flow characteristics were noted to affect the hot-spot temperatures and the flow uniformity. The laminarization of the flow in the near-boundary regions in the coolant channels was noted to have a strong link with the local increase of the hot-spot temperatures. The primary and secondary flow structures were both noted to impact the flow uniformity. This

information could possibly be used for improved design of heat exchangers by e.g. flow control techniques.

5. Conclusions

A liquid cooling heat exchanger for the cooling of high power electronics was numerically and experimentally analysed. The numerical analysis was based on CFD simulations using a CHT solver. For experimental testing, the heat exchanger was 3D printed out of aluminium. The simulations were executed with two different steady state RANS simulations, the standard $k-\epsilon$ model and the $k-\omega$ SST model, in addition to LES.

The standard $k-\epsilon$ simulations revealed lower surface temperature levels and absence of various small scale flow structures in comparison to $k-\omega$ SST and LES simulations. As a general finding, $k-\omega$ SST and LES models provided consistent results. The flow characteristics, such as flow separation and primary and secondary vortices, were noted to affect the uniformity of the flow rate between the channels. The transitional flow character (laminarization) inside the channels was noted to increase the simulated hot-spot temperatures. $k-\omega$ SST was used as the base case for mass flow rate assessment, results of which indicated independence of the inlet Reynolds number to the velocity distribution

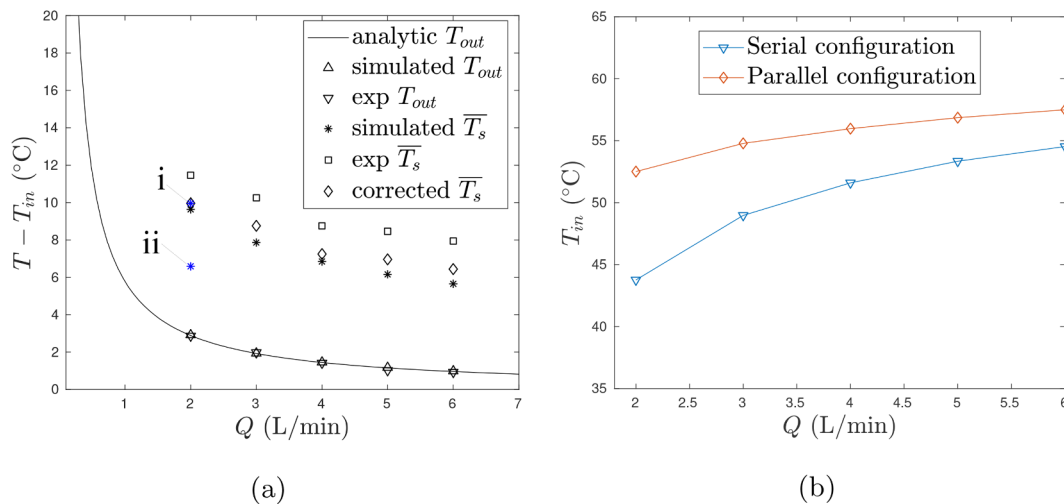


Fig. 14. (a): The outlet temperature and the surface temperatures referenced to the inlet temperature as a function of the mass flow rate. (i): LES, (ii): $k-\epsilon$. The figure reveals similar surface temperature levels between the simulations and the corrected experimental values. The surface temperature calculated with $k-\epsilon$ model is highly underestimated. (b): The highest allowed inlet temperatures to ensure proper cooling.

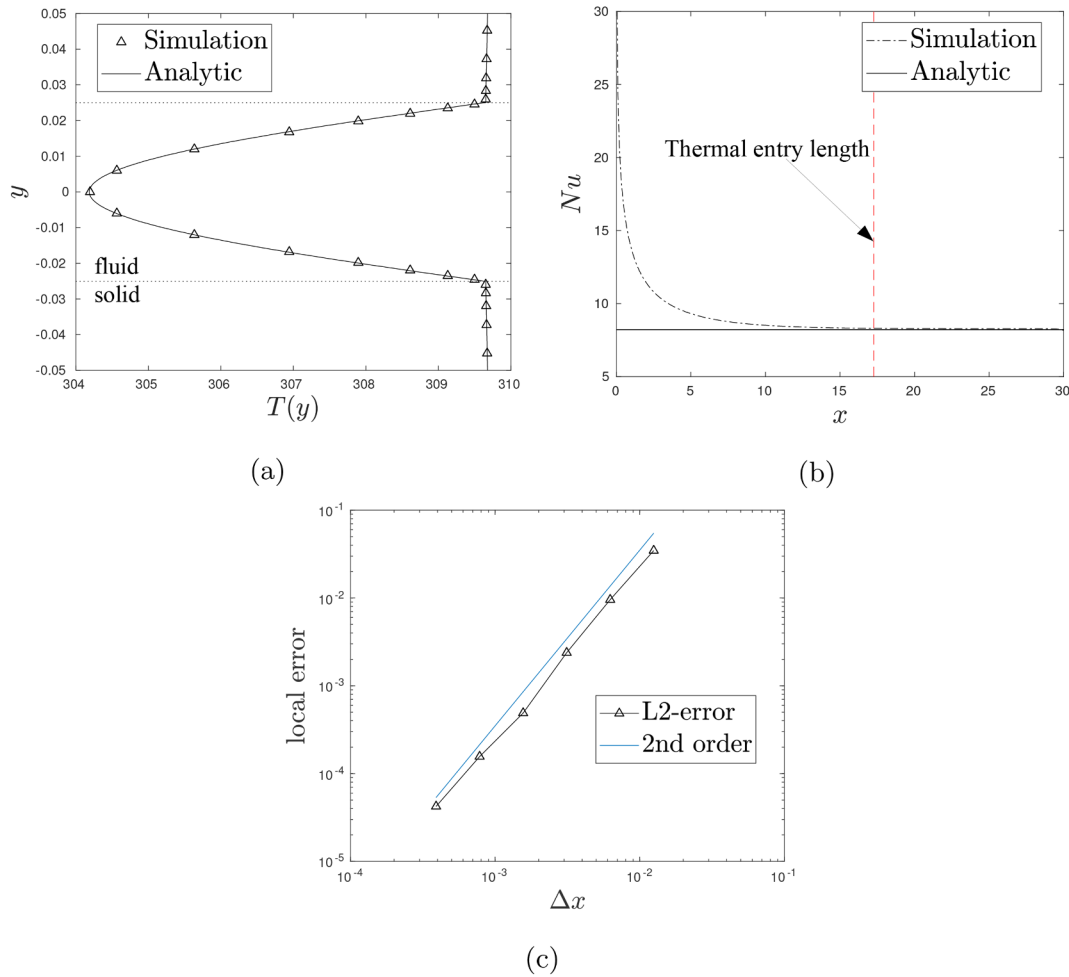


Fig. A.15. (a): Temperature profile in the fully developed region from simulations compared to the analytic solution with a maximum local error of $e = 10^{-3}$. (b): The Nusselt number as a function of the stream wise coordinate compared to the analytic solution (c): The mesh convergence of the L2 error.

in the channels.

The experimental and numerical results revealed that the present design is functional for its cooling purpose. With a power of 400 W, a flow rate of 2 L/min and a 27 °C inlet temperature, the highest surface temperatures were below 40 °C with all the models, which was also verified by the experiments. There were no significant differences in the temperature levels in the $k-\omega$ SST and LES compared to the experiments. The present numerical framework has potential to be used in quantitative analysis of various convective cooling and heat transfer applications.

CRediT authorship contribution statement

Alpo Laitinen: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing - original draft, Visualization. **Kari Saari:** Investigation, Resources. **Kirsi Kukko:** Methodology, Resources. **Petteri Peltonen:** Writing - review & editing.

Appendix A. Solver verification for laminar flow

The purpose of this Appendix is to validate the CHT solver for 2D laminar channel flow case in steady conditions for the present ILES method. The present CHT solver is verified against an analytic solution of a fully developed steady state 2D laminar flow between heated parallel plates. The verification case was selected due to the availability of an analytic solution and the similarities between the test case to the heat transfer configuration of the present study. The inflow in the present study is turbulent, however, the flow laminarizes in the heat transfer channels, making the laminar channel flow a relevant verification case.

The Reynolds number between the plates is $Re = 1040$, with relevance to the 3D heat exchanger case. In the 2D simulations the number of grid points in the y-direction is 15 in the fluid region with refinement near the boundaries, and 5 in both solid regions. In the fully developed laminar

Erkki Laurila: Writing - review & editing. **Jouni Partanen:** Supervision. **Ville Vuorinen:** Supervision, Project administration, Conceptualization, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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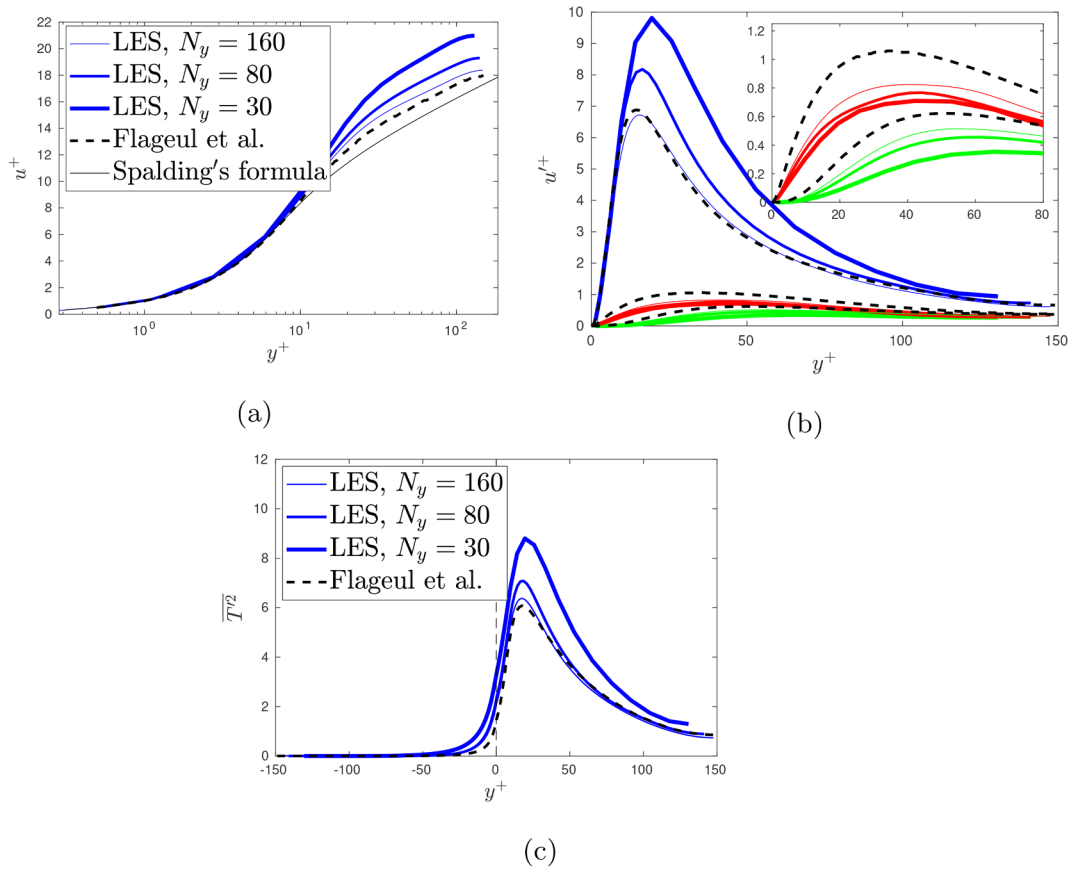


Fig. B.16. (a): The dimensionless velocity compared to Spalding's formula and DNS data. (b): The Reynolds stresses compared to DNS data. Blue line: u_{xx} , green line: u_{yy} , red line: u_{zz} . (c): Temperature fluctuations compared to DNS data. Dashed vertical line represents the fluid–solid interface. $y^+ < 0$ is in the solid domain and $y^+ > 0$ in the fluid domain. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

regime, the Nusselt number with constant heat flux boundary condition is known from the literature to be $Nu = 8.23$ (Shah and London, 1971). The thermal entrance length is estimated as

$$l_{th} \approx 0.05 Re D_h Pr \quad (16)$$

where Re is the Reynolds number, Pr is the Prandtl number and D_h is the hydraulic diameter (Bergman et al., 2011). The numerical solution is compared with the analytic solution in the fully developed region.

The temperature field in the fully developed region in the simulations is shown in Fig. A.15a along with the analytic solution. Additionally, a Nusselt number comparison is provided in Fig. A.15b. The numerical results are noted to be in excellent agreement with the analytical solutions. Furthermore, the L2 error with 6 different meshes is shown in Fig. A.15c, confirming second order spatial accuracy of the numerical solver. The mesh convergence study is performed with uniform cell size in y-direction, i.e. no refinement near the boundaries was done. The presented verification case provides numerical evidence on the functionality of the present CHT approach in steady state conditions, hence motivating heat exchanger design in the scope of the present paper.

Appendix B. Solver verification for turbulent flow

The purpose of this Appendix is to verify the CHT solver for 3D turbulent flow for the present ILES method against DNS data (Flageul et al., 2017). A 3D turbulent channel flow is carried out using the present ILES with $\gamma = 0.2$ at friction Reynolds number of $Re_\tau = 150$. As confirmed by the L2-error analysis in A, the present solver is 2nd order accurate in space. The height of the channel is 2δ , the thickness of the solid is $\delta_s = \delta$ and the length of the channel is $L = 8\delta$. The lateral boundaries of the channel are periodic and the domain width in the lateral direction is 2δ . The case is simulated with three different mesh resolutions. The coarsest grid resolution is $N_f = (N_x, N_y, N_z) = (40, 30, 20)$ for the fluid domain and $N_s = (30, 50, 20)$ for the solid domain. Similarly, the medium grid resolution is $N_f = (80, 40, 40)$ and $N_s = (80, 50, 40)$, while the finest grid resolution is $N_f = (120, 160, 80)$ and $N_s = (120, 50, 80)$. The number of cells in x and z directions are the same in the fluid and solid domains and the number of cells in y-direction in the solid is the same in each simulation. In the present heat exchanger study the flow laminarizes in-between the channels so the flow situations are not the same as in this benchmark case. However, the value of the near-wall y^+ is similar.

For velocity and temperature, the data is mapped from a recycling plane at $x = 0.75L$ back to the inlet. At the fluid–solid boundary, velocity obeys the no-slip condition. A constant heat flux is imposed at the top and bottom parts of the solid plates. A heat sink is applied in the fluid domain following (Flageul et al. (2019)). The computational set-up allows simulation of hydrodynamically and thermally fully developed flow conditions. The thermal properties of the fluid and solid are chosen based on the available DNS data such that the ratio of the thermal diffusivities is $G = \alpha_f/\alpha_s = 2$ and the ratio of the thermal conductivities is $G_2 = \kappa_s/\kappa_f = 0.5$.

The dimensionless velocity profile is plotted in Fig. B.16a. The velocity profile is plotted along with DNS and Spalding's formula (Mandal and

Mazumdar, 2015). The Reynolds stresses are plotted in Fig. B.16b and compared to DNS. The velocity profile and the Reynolds stresses are non-dimensionalized with a friction velocity $u_\tau = \frac{\nu Re_\tau}{\delta}$. The temperature fluctuations are plotted in Fig. B.16c and compared to DNS data. The temperature fluctuations are non-dimensionalized using a friction temperature $T_\tau = \frac{Cf q_w}{k_f u_\tau}$. The results for the velocity, velocity fluctuations and the temperature fluctuations with all the meshes show relatively good correspondence with the DNS.

In summary of Appendices A and B, the following conclusions can be made on the present CHT approach. Firstly, the steady state laminar temperature profile is in excellent agreement with the analytic solution and the solver is 2nd order accurate in space. Herein, this is of key importance since most heat transfer occurs in low Reynolds number regime. Secondly, the turbulent velocity and temperature statistics are in agreement with DNS. Laminar and weakly turbulent flow cases with heat transfer are highly relevant with respect to the present application. We remind that the present ILES approach and the $k-\omega$ SST approach provide very similar temperature profiles in the actual heat exchanger geometry (see Fig. 10c).

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